

Programming Languages -- Guided Notes

Unit tp-3.7 | CompTIA Network+ N10-009 Obj. FC0-U71 3.7 | N10-008 FC0-U71 3.7

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A programming language is a formal language used to write _____ that computers can execute. Source code is written in _____ -readable text, then converted to machine code. The two main conversion methods are _____ (converts all code at once before running) and _____ (converts and runs line by line).

Question 2

_____ code (binary machine instructions) is what the CPU executes directly. _____ language uses mnemonics (like MOV, ADD) to represent machine instructions. High-level languages like Python and Java are _____ readable and must be compiled or interpreted before the CPU can run them.

Question 3

Python is an _____ language known for readable syntax and use in data science, automation, and web development. Java is a _____ language that compiles to bytecode run on the _____ (JVM), enabling write-once run-anywhere portability. JavaScript runs primarily in _____ to make web pages interactive.

Question 4

C and C++ are _____ -level languages often used for _____ programming, game engines, and performance-critical applications. They give programmers direct access to _____ management, which increases performance but also increases the risk of _____ vulnerabilities like buffer overflows.

Question 5

SQL (Structured Query Language) is used to _____ and _____ data in relational databases. HTML (Hypertext Markup Language) defines the _____ of web pages. CSS (Cascading Style Sheets) defines the _____ (colors, fonts, layout). Neither HTML nor CSS is technically a _____ language -- they are markup/styling languages.

Question 6

A scripting language (Python, Bash, PowerShell) is typically _____ and used for _____ tasks. A PowerShell script can automate _____ administrative tasks like creating user accounts. A Bash script on Linux can automate _____ and file operations. Scripts reduce _____ compared to doing tasks manually.

Question 7

An IDE (Integrated Development Environment) combines a _____ editor, _____ (finds syntax errors), _____ (automates build/run), and _____ (steps through code to find logical errors) into one tool. Popular IDEs include Visual Studio Code, PyCharm, and _____.

Question 8 -- Real World Application

REAL WORLD: A student wants to build a personal website, create some scripts to automate their homework file organization, and eventually get into data analysis. They ask you which programming language they should learn first. Recommend one language, give three reasons it is the best starting point for their goals, and name one limitation.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-3-7-programming-languages/  
tp-3-7-programming-languages-answer-key.pdf
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